*For the use only of a Registered Medical Practitioner*

PRESCRIBING INFORMATION

# ROSART TABLETS

## (Losartan Potassium Tablets 50 / 100 mg)

## COMPOSITION

ROSART TABLETS 50 mg
Each Film coated Tablet contains
Losartan Potassium…………….50 mg

ROSART TABLETS 100 mg
Each Film coated Tablet contains
Losartan Potassium…………….100 mg

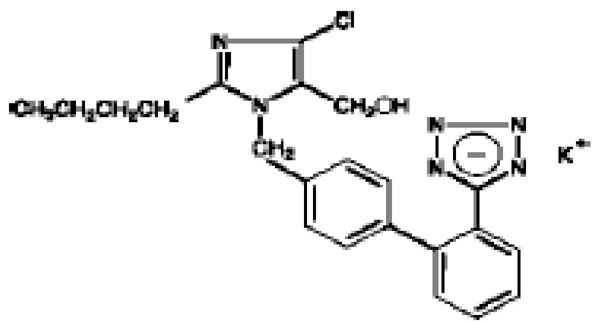
## PRODUCT DESCRIPTION

ROSART TABLETS 50 mg
White to off white, circular, biconvex, film-coated tablets with score line on one side and plain on the other side. Diameter 7.6 mm

ROSART TABLETS 100 mg
White to off white, circular, biconvex, film-coated tablets with score line on one side and plain on the other side. Diameter 9.6 mm

## DESCRIPTION

ROSART (Losartan potassium) is an angiotensin II receptor (type $AT_1$) antagonist. Losartan potassium, a non-peptide molecule, is chemically described as 2-butyl-4-chloro-1-[p-(o-1*H*-tetrazol-5-ylphenyl)benzyl]imidazole-5-methanol monopotassium salt. Its empirical formula is $C_{22}H_{22}ClKN_6O$ and molecular weight is 461.01. The structural formula is as below:



LOSARTAN POTASSIUM

Oxidation of the 5-hydroxymethyl group on the imidazole ring results in the active metabolite of losartan.

## PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

### *Therapeutic Class*

Losartan potassium, the first of a new class of agents for the treatment of hypertension, is an angiotensin II receptor (type AT1) antagonist. Losartan also provides a reduction in the combined risk of cardiovascular death, stroke, and myocardial infarction in hypertensive patients with left ventricular hypertrophy and provides renal protection for type 2 diabetic patients with proteinuria.

### *Mechanism of Action*

Angiotensin II, a potent vasoconstrictor, is the primary active hormone of the renin-angiotensin system, and a major determinant of the pathophysiology of hypertension. Angiotensin II binds to the $AT_1$ receptor found in many tissues (e.g., vascular smooth muscle, adrenal gland, kidneys, and the heart) and elicits several important biological actions, including vasoconstriction and the release of aldosterone. Angiotensin II also stimulates smooth muscle cell proliferation. A second angiotensin II receptor has been identified as the $AT_2$ receptor subtype, but it plays no known role in cardiovascular homeostasis.

Losartan is a potent, synthetic, orally active compound. It binds selectively to the $AT_1$ receptor. *In vitro* and *in vivo*, both losartan and its pharmacologically active carboxylic acid metabolite (E-3174) block all physiologically relevant actions of angiotensin II, regardless of the source or route of synthesis. In contrast to some peptide antagonists of angiotensin II, losartan has no agonist effects.

Losartan binds selectively to the $AT_1$ receptor and does not bind to or block other hormone receptors or ion channels important in cardiovascular regulation. Furthermore, losartan does not inhibit ACE (kininase II), the enzyme that degrades bradykinin. Consequently, effects not directly related to blocking the AT1 receptor, such as the potentiation of bradykinin-mediated effects or the generation of edema (losartan 1.7%, placebo 1.9%), has not been reported with losartan.

- Pharmacodynamics

Losartan inhibits systolic and diastolic pressor responses to angiotensin II infusions. At peak, 100 mg of losartan potassium inhibits these responses by approximately 85%; 24 hours after single and multiple-dose administration, inhibition is about 26-39%.

During losartan administration, removal of angiotensin II negative feedback on renin secretion leads to increased plasma renin activity. Increases in plasma renin activity lead to increases in angiotensin II in plasma. During chronic treatment of hypertensive patients with 100 mg/day losartan, approximately 2-3 fold increases of plasma angiotensin II has been reported at time of peak plasma drug concentrations. In some patients, greater increases has been reported, particularly during short term treatment. However, antihypertensive activity and suppression of plasma aldosterone concentration were apparent at 2 and 6 weeks, indicating effective angiotensin II receptor blockade. After discontinuation of losartan, plasma renin activity and angiotensin II levels declined to untreated levels within 3 days.

Since losartan is a specific antagonist of the angiotensin II receptor type $AT_1$, it does not inhibit ACE (kininase II), the enzyme that degrades bradykinin. The effects of 20 mg and 100 mg of losartan potassium and an ACE inhibitor on responses to angiotensin I, angiotensin II and bradykinin, losartan has been reported to block responses to angiotensin I and angiotensin II without affecting responses to bradykinin. This finding is consistent with losartan's specific mechanism of action. In contrast, the ACE inhibitor has been reported to block responses to angiotensin I and enhance responses to bradykinin without altering the response to angiotensin II, thus providing a pharmacodynamic distinction between losartan and ACE inhibitors.

Plasma concentrations of losartan and its active metabolite and the antihypertensive effect of losartan increase with increasing dose. Since losartan and its active metabolite are both angiotensin II receptor antagonists, they both contribute to the antihypertensive effect.

In normal males, the administration of 100 mg of losartan potassium, under dietary high- and low-salt conditions, did not alter glomerular filtration rate, effective renal plasma flow or filtration fraction. Losartan had a natriuretic effect which has been more pronounced on a low-salt diet and did not appear to be related to inhibition of early proximal reabsorption of sodium. Losartan also caused a transient increase in urinary uric acid excretion.

In non-diabetic hypertensive patients with proteinuria (≥ 2 g/24 hours) treated for 8 weeks, the administration of losartan potassium 50 mg titrated to 100 mg significantly reduced proteinuria by 42%. Fractional excretion of albumin and IgG also was significantly reduced. In these patients, it has reported that losartan maintained glomerular filtration rate and reduced filtration fraction.

In postmenopausal hypertensive women treated for 4 weeks, 50 mg of losartan potassium had no effect on renal or systemic prostaglandin levels.

Losartan has no effect on autonomic reflexes and no sustained effect on plasma norepinephrine.

It has been reported that, Losartan potassium, administered in doses of up to 150 mg once daily, did not cause clinically important changes in fasting triglycerides, total cholesterol or HDL-cholesterol in patients with hypertension. The same doses of losartan has no effect on fasting glucose levels.

Generally losartan causes a decrease in serum uric acid (usually <0.4 mg/dL) which has been reported to be persistent in chronic therapy

- Pharmacokinetic

*Absorption*

Following oral administration, losartan is well absorbed and undergoes first-pass metabolism, forming an active carboxylic acid metabolite and other inactive metabolites. The systemic bioavailability of losartan tablets is approximately 33%. Mean peak concentrations of losartan and its active metabolite are reached in 1 hour and in 3-4 hours, respectively. No clinically significant effect on the plasma concentration profile of losartan has been reported when administered with a standardized meal.

*Distribution*

Both losartan and its active metabolite are ≥ 99% bound to plasma proteins, primarily albumin. The volume of distribution of losartan is 34 L. Losartan has been reported to crosse the blood-brain barrier poorly in rats, if at all.

*Metabolism*

About 14% of an IV or orally-administered dose of losartan is converted to its active metabolite. Following oral and intravenous administration of potassium, circulating plasma radioactivity primarily is attributed to losartan and its active metabolite. Minimal conversion of losartan to its active metabolite has been reported.

In addition to the active metabolite, inactive metabolites are formed, including 2 major metabolites formed by hydroxylation of the butyl side chain and a minor metabolite, an N-2 tetrazole glucuronide.

*Excretion*

Plasma clearance of losartan and its active metabolite is about 600 mL/min and 50 mL/min, respectively. Renal clearance of losartan and its active metabolite is about 74 mL/min and 26 mL/min, respectively. After oral administration of losartan, about 4% of the dose is excreted unchanged in the urine, and about 6% of the dose is excreted in the urine as active metabolite. The pharmacokinetics of losartan and its active metabolite are linear with oral losartan potassium doses up to 200 mg.

After oral administration, plasma concentrations of losartan and its active metabolite decline poly-exponentially with a terminal half-life of about 2 hours and 6-9 hours, respectively. During once-daily dosing with 100 mg, neither losartan nor its active metabolite accumulates significantly in plasma.

Both biliary and urinary excretion contribute to the elimination of losartan and its metabolites. Following an oral dose of $^{14}C$-labeled losartan in man, about 35% of radioactivity has been reported to be recovered in the urine and 58% in the feces. Following an intravenous dose of $^{14}C$-labeled losartan in man, about 43% of radioactivity has been reported to be recovered in the urine and 50% in the feces.

*Characteristics in Patients*

The plasma concentrations of losartan and its active metabolite reported in elderly male hypertensives are not significantly different from those reported in young male hypertensives.

Plasma concentrations of losartan has been reported up to 2-fold higher in female hypertensives as compared to male hypertensives. Concentrations of the active metabolite has not been reported different in males and females. This apparent pharmacokinetic difference is not judged to be of clinical significance.

Following oral administration in patients with mild to moderate alcoholic cirrhosis of the liver, plasma concentrations of losartan and its active metabolite were, respectively, 5-fold and 1.7-fold greater than those reported in young male volunteers.

Plasma concentrations of losartan are not altered in patients with creatinine clearance above 10 ml/min. Compared to patients with normal renal function, the AUC for losartan is approximately 2-fold greater in haemodialysis patients. Plasma concentrations of the active metabolite are not altered in patients with renal impairment or in haemodialysis patients. Neither losartan nor the active metabolite can be removed by haemodialysis.

## INDICATIONS

*Hypertension*

ROSART is indicated for the treatment of hypertension.

*Hypertensive Patients with Left Ventricular Hypertrophy*
ROSART is indicated to reduce the risk of stroke in patients with hypertension and left ventricular hypertrophy, but there is evidence that this benefit does not apply to Black patients.

*Nephropathy in Type 2 Diabetic Patients*
ROSART is indicated for the treatment of diabetic nephropathy with an elevated serum creatinine and proteinuria (urinary albumin to creatinine ratio ≥ 300 mg/g) in patients with type 2 diabetes and history of hypertension. In this population, ROSART reduces the rate of progression of nephropathy as measured by the occurrence of doubling the serum creatinine or end-stage renal disease (need for dialysis or renal transplantation) or death.

## DOSE AND METHOD OF ADMINISTRATION

ROSART may be administered with or without food.

ROSART may be administered with other antihypertensive agents.

### Hypertension
The usual starting and maintenance dose is 50 mg once daily for most patients. The maximal antihypertensive effect is attained 3-6 weeks after initiation of therapy. Some patients may receive an additional benefit by increasing the dose to 100 mg once daily.

For patients with intravascular volume-depletion (e.g., those treated with high-dose diuretics), a starting dose of 25 mg once daily should be considered (see WARNINGS AND PRECAUTIONS).

No initial dosage adjustment is necessary for elderly patients or for patients with renal impairment, including patients on dialysis. A lower dose should be considered for patients with a history of hepatic impairment (see WARNINGS AND PRECAUTIONS).

Hypertensive Patients with Left Ventricular Hypertrophy

The usual starting dose is 50 mg of ROSART once daily. A low dose of hydrochlorothiazide should be added and/or the dose of ROSART should be increased 100 mg once daily based on blood pressure response.

Renal Protection in Type 2 Diabetic Patients with Proteinuria and Hypertension

The usual starting dose is 50 mg once daily. The dose may be increased to 100 mg once daily based on blood pressure response. ROSART may be administered with other antihypertensive agents (eg, diuretics, calcium-channel blockers, α- or β-blockers and centrally acting agents) as well as with insulin and other commonly used hypoglycemic agents (eg, sulfonylureas, glitazones and glucosidase inhibitors).

*Use in Special Populations*

*Pediatric Use*

*Neonates with a history of in utero exposure to losartan:*

If oliguria or hypotension occur, direct attention toward support of blood pressure and renal perfusion. Exchange transfusions or dialysis may be required as a means of reversing hypotension and/or substituting for disordered renal function.

Antihypertensive effects of losartan have been reported in hypertensive pediatric patients aged > 1 month to 16 years.

The active metabolite is formed from losartan in all age groups. Pharmacokinetics of losartan and its active metabolite has been reported to be generally similar across the > 1 month to 16 years age groups and consistent with pharmacokinetic historic data in adults.

Losartan administration once daily has been reported to lower trough blood pressure in a dose-dependent manner in pediatric patients 6 to 16 years of age. The dose response to losartan has been reported across all subgroups (e.g., age, Tanner stage, gender, race). However, the lowest doses reported, 2.5 mg and 5 mg, corresponding to an average daily dose of 0.07 mg/kg, did not appear to offer consistent antihypertensive efficacy.

For patients who can swallow tablets, the recommended dose is 25 mg once daily in patients ≥20 to <50 kg. The dose can be increased to a maximum of 50 mg once daily. In patients >50 kg, the starting dose is 50 mg once daily. The dose can be increased to a maximum of 100 mg once daily.

In pediatric patients who are intravascularly volume depleted, these conditions should be corrected prior to administration of losartan.

The adverse experience profile for pediatric patients appears to be similar to that reported in adult patients.

Losartan is not recommended in pediatric patients with glomerular filtration rate <30 mL/min/1.73 $m^2$, in pediatric patients with hepatic impairment, or in neonates as no data are available.

## CONTRAINDICATIONS

ROSART TABLETS is contraindicated in patients who are hypersensitive to any component of this product.

ROSART TABLETS should not be administered with aliskiren in patients with diabetes (see DRUG INTERACTIONS).

ROSART TABLETS is contraindicated in $2^{nd}$ and $3^{rd}$ trimester of pregnancy.

ROSART TABLETS is contraindicated in severe hepatic impairment.

## WARNINGS AND PRECAUTIONS

*Fetal Toxicity*
Use of drugs that act on the renin-angiotensin system during the second and third trimesters of pregnancy reduces fetal renal function and increases fetal and neonatal morbidity and death. Resulting oligohydramnios can be associated with fetal lung hypoplasia and skeletal deformations. Potential neonatal adverse effects include skull hypoplasia, anuria, hypotension, renal failure, and death. When pregnancy is detected, discontinue losartan as soon as possible. See USE IN SPECIAL POPULATIONS

*Hypersensitivity*
Angioedema. (See SIDE EFFECTS/UNDESIRABLE EFFECTS)

*Hypotension and electrolyte/fluid imbalance*
In patients who are intravascularly volume-depleted (e.g., those treated with high-dose diuretics) symptomatic hypotension may occur. These conditions should be corrected prior to administration of losartan potassium, or a lower starting dose should be used. This also applies to children 6 to 18 years of age.

Electrolyte imbalances are common in patients with renal impairment, with or without diabetes, and should be addressed. In type 2 diabetic patients with proteinuria, the incidence of hyperkalaemia has been reported to be higher in the group treated with losartan as compared to the placebo group.

Concomitant use of other drugs that may increase serum potassium may lead to hyperkalemia (see DRUG INTERACTIONS).

*Liver function impairment*
Significantly increased plasma concentrations of losartan has been reported in cirrhotic patients. Hence, in patients with history of hepatic impairment a lower dose should be considered. There is no therapeutic experience with losartan in patients with severe hepatic impairment. Therefore losartan must not be

administered in patients with severe hepatic impairment. see DOSAGE AND ADMINISTRATION and PHARMACOLOGY, Pharmacokinetic).

Losartan is not recommended in children with hepatic impairment.

*Renal function impairment*
As a consequence of inhibiting the renin-angiotensin system, changes in renal function including renal failure have been reported. In susceptible individuals; these changes in renal function may be reversible upon discontinuation of therapy.

As with other drugs that affect the renin-angiotensin- system, increases in blood urea and serum creatinine have also been reported in patients with bilateral renal artery stenosis or stenosis of the artery to a solitary kidney; these changes in renal function may be reversible upon discontinuation of therapy.

*Use in Paediatric patients with renal impairment*
Losartan is not recommended in children with glomerular filtration rate < 30 ml/min/1.73 $m^2$ as no data has been reported.

Renal function should be regularly monitored during treatment with losartan as it may deteriorate. This applies particularly when losartan is given in the presence of other conditions (fever, dehydration) likely to impair renal function.

Concomitant use of losartan and ACE-inhibitors has shown to impair renal function. Therefore, concomitant use is not recommended.

*Renal transplantation*
No data has been reported in patients with recent kidney transplantation.

*Primary hyperaldosteronism*
Patients with primary aldosteronism generally will not respond to antihypertensive medicinal products acting through inhibition of the renin-angiotensin system. Therefore, the use of losartan is not recommended.

*Coronary heart disease and cerebrovascular disease*
As with any antihypertensive agents, excessive blood pressure decrease in patients with ischaemic cardiovascular and cerebrovascular disease could result in a myocardial infarction or stroke.

*Heart failure*
In patients with heart failure, with or without renal impairment, there is - as with other medicinal products acting on the renin-angiotensin system - a risk of severe arterial hypotension, and (often acute) renal impairment.

There is no sufficient therapeutic data with losartan in patients with heart failure and concomitant severe renal impairment, in patients with severe heart failure (NYHA class IV) as well as in patients with heart failure and symptomatic life-threatening cardiac arrhythmias. Therefore, losartan should be used with caution in these patient groups. The combination of losartan with a beta-blocker should be used with caution.

*Aortic and mitral valve stenosis, obstructive hypertrophic cardiomyopathy*
As with other vasodilators, special caution is indicated in patients suffering from aortic or mitral stenosis, or obstructive hypertrophic cardiomyopathy.

Pregnancy
Losartan should not be initiated during pregnancy. Unless continued losartan therapy is considered essential, patients planning pregnancy should be changed to alternative anti-hypertensive treatments which have an established safety profile for use in pregnancy. When pregnancy is diagnosed, treatment with losartan should be stopped immediately, and, if appropriate, alternative therapy should be started.

*Dual blockade of the renin-angiotensin-aldosterone system (RAAS)*
The concomitant use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren increases the risk of hypotension, hyperkalaemia, and decreased renal function (including acute renal failure). Dual blockade of RAAS through the combined use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren is therefore not recommended.

If dual blockade therapy is considered absolutely necessary, this should only occur under specialist supervision and subject to frequent close monitoring of renal function, electrolytes and blood pressure.

ACE-inhibitors and angiotensin II receptor blockers should not be used concomitantly in patients with diabetic nephropathy.

*Lactose intolerance*
ROSART contain lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency, or glucose-galactose malabsorption should not take this medicine.

*Effects on ability to drive and use machines*
There is no data to suggest that losartan affects the ability to drive and use machines.

USE IN SPECIAL POPULATIONS

- Pregnancy

Although there is no data been reported with the use of losartan in pregnant women, animal studies with losartan potassium have reported fetal and neonatal injury and death, the mechanism of which is believed to be pharmacologically mediated through effects on the renin-angiotensin system. In humans, fetal renal perfusion, which is dependent upon the development of the renin-angiotensin system, begins in the second trimester; thus, risk to the fetus increases if losartan is administered during the second or third trimesters of pregnancy.

Use of drugs that act on the renin-angiotensin system during the second and third trimesters of pregnancy reduces fetal renal function and increases fetal and neonatal morbidity and death. Resulting oligohydramnios can be associated with fetal lung hypoplasia and skeletal deformations. Potential neonatal adverse effects include skull hypoplasia, anuria, hypotension, renal failure, and death. When pregnancy is detected, discontinue losartan as soon as possible.

These adverse outcomes are usually associated with the use of these drugs in the second and third trimesters of pregnancy. Most reported data examining fetal abnormalities after exposure to antihypertensive use in the first trimester have not distinguished drugs affecting the renin-angiotensin system from other antihypertensive agents. Appropriate management of maternal hypertension during pregnancy is important to optimize outcomes for both mother and fetus.

In the unusual case that there is no appropriate alternative to therapy with drugs affecting the renin-angiotensin system for a particular patient, apprise the mother of the potential risk to the fetus. Perform serial ultrasound examinations to assess the intra-amniotic environment. If oligohydramnios is observed, discontinue losartan, unless it is considered life-saving for the mother. Fetal testing may be appropriate, based on the week of pregnancy. Patients and physicians should be aware, however, that oligohydramnios may not appear until after the fetus has sustained irreversible injury. Closely observe infants with histories of in utero exposure to losartan for hypotension, oliguria, and hyperkalemia.

- Lactation

It is not known whether losartan is excreted in human milk. Because it is reported that, many drugs are excreted in human milk and because of the potential for adverse effects on the nursing infant, a decision should be made whether to discontinue nursing or discontinue the drug, taking into account the importance of the drug to the mother.

- Elderly

No age-related difference in the efficacy or safety profile of losartan has been reported.

- Race

As reported, the benefits of losartan on cardiovascular morbidity and mortality compared to atenolol do not apply to Black patients with hypertension and left ventricular hypertrophy although both treatment regimens effectively lowered blood pressure in Black patients.

## DRUG INTERACTIONS

No drug interactions of clinical significance have been reported with hydrochlorothiazide, digoxin, warfarin, cimetidine, phenobarbital, ketoconazole, and erythromycin. Rifampin and fluconazole have been reported to reduce levels of active metabolite. The clinical consequences of these interactions have not been reported.

As with other drugs that block angiotensin II or its effects, concomitant use of potassium-sparing diuretics (e.g., spironolactone, triamterene, amiloride), potassium supplements, salt substitutes containing potassium, or other drugs that may increase serum potassium (e.g., trimethoprim-containing products) may lead to increases in serum potassium.

*Lithium*

As with other drugs which affect the excretion of sodium lithium’s excretion may be reduced. Serum lithium levels should therefore be monitored carefully if lithium salts are to be co-administered with angiotensin II receptor antagonists.

Non-steroidal anti-inflammatory drugs (NSAIDs) including selective cyclooxygenase-2 inhibitors (COX-2 inhibitors) may reduce the effect of diuretics and other antihypertensive drugs. Therefore, the antihypertensive effect of angiotensin II receptor antagonists or ACE inhibitors may be attenuated by NSAIDs including selective COX-2 inhibitors.

In some patients with compromised renal function (e.g., elderly patients or patients who are volume-depleted, including those on diuretic therapy) who are being treated with non-steroidal anti-inflammatory drugs, including selective cyclooxygenase-2 inhibitors, the co-administration of angiotensin II receptor antagonists or ACE inhibitors may result in a further deterioration of renal function, including possible acute renal failure. These effects are usually reversible. Therefore, the combination should be administered with caution in patients with compromised renal function.

Dual blockade of the renin-angiotensin-aldosterone system (RAAS) with angiotensin receptor blockers, ACE inhibitors, or aliskiren is associated with increased risks of hypotension, syncope, hyperkalemia, and changes in renal function (including acute renal failure) compared to monotherapy. Close monitor blood pressure, renal function, and electrolytes in patients on losartan and other agents that affect the RAAS. Do not co-administer aliskiren with losartan in patients with diabetes. Avoid use of aliskiren with losartan in patients with renal impairment (GFR <60 ml/min).

Other antihypertensive agents may increase the hypotensive action of losartan. Concomitant use with other substances which may induce hypotension as an adverse reaction (like tricyclic antidepressants, antipsychotics, baclofen and amifostine) may increase the risk of hypotension.

Losartan is predominantly metabolised by cytochrome P450 (CYP) 2C9 to the active carboxy-acid metabolite. It has been reported that fluconazole (inhibitor of CYP2C9) decreases the exposure to the active metabolite by approximately 50%. It has been reported that concomitant treatment of losartan with rifampicin (inducer of metabolism enzymes) gave a 40% reduction in plasma concentration of the active metabolite. The clinical relevance of this effect is unknown. No difference in exposure has been reported with concomitant treatment with fluvastatin (weak inhibitor of CYP2C9).

## SIDE EFFECTS/UNDESIRABLE EFFECTS

The most common adverse event was dizziness.

Table The frequency of adverse reactions reported with losartan

| Adverse reaction | Frequency of adverse reaction by indication | | | | Other |
|---|---|---|---|---|---|
| | Hypertension | Hypertensive patients with left-ventricular hypertrophy | Chronic Heart Failure | Hypertension and type 2 diabetes with renal disease | Post-marketing experience |
| <u>Blood and lymphatic system disorders</u> | | | | | |
| anaemia | | | common | | frequency not known |

| | | | | | |
|---|---|---|---|---|---|
| thrombocytopenia | | | | | frequency not known |
| **Immune system disorders** | | | | | |
| hypersensitivity reactions, anaphylactic reactions, angioedema*, and vasculitis** | | | | | rare |
| **Psychiatric disorders** | | | | | |
| depression | | | | | frequency not known |
| **Nervous system disorders** | | | | | |
| dizziness | common | common | common | common | |
| somnolence | uncommon | | | | |
| headache | uncommon | | uncommon | | |
| sleep disorders | uncommon | | | | |
| paraesthesia | | | rare | | |
| migraine | | | | | frequency not known |
| dysgeusia | | | | | frequency not known |
| **Ear and labyrinth disorders** | | | | | |
| vertigo | common | common | | | |
| tinnitus | | | | | frequency not known |
| **Cardiac disorders** | | | | | |
| palpitations, tachycardia | uncommon | | | | |
| angina pectoris | uncommon | | | | |
| syncope | | | rare | | |
| atrial fibrillation | | | rare | | |
| cerebrovascular accident | | | rare | | |
| **Vascular disorders** | | | | | |
| (orthostatic) hypotension (including dose- related orthostatic effects) ‖ | uncommon | | common | common | |

| | | | | | |
|---|---|---|---|---|---|
| Respiratory, thoracic and mediastinal disorders | | | | | |
| dyspnea | | | uncommon | | |
| Cough, Nasal congestion, Pharyngitis, Sinus disorder, Upper respiratory infection | | | uncommon | | frequency not known |
| Gastrointestinal disorders | | | | | |
| abdominal pain | uncommon | | | | |
| obstipation | uncommon | | | | |
| diarrhoea | | | uncommon | | frequency not known |
| nausea | | | uncommon | | |
| vomiting | | | uncommon | | |
| Hepatobiliary disorders | | | | | |
| pancreatitis | | | | | frequency not known |
| hepatitis | | | | | rare |
| liver function abnormalities | | | | | frequency not known |
| Skin and subcutaneous tissue disorders | | | | | |
| urticaria | | | uncommon | | frequency not known |
| pruritus | | | uncommon | | frequency not known |
| rash | uncommon | | uncommon | | frequency not known |
| photosensitivity | | | | | frequency not known |
| Musculoskeletal and connective tissue disorders | | | | | |
| Myalgia | | | | | frequency not known |
| arthralgia | | | | | frequency not known |
| rhabdomyolysis | | | | | frequency not known |
| Renal and urinary disorders | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| renal impairment | | | common | | |
| renal failure | | | common | | |
| Reproductive system and breast disorders | | | | | |
| erectile dysfunction / impotence | | | | | frequency not known |
| General disorders and administration site conditions | | | | | |
| asthenia | uncommon | common | uncommon | common | |
| fatigue | uncommon | common | uncommon | common | |
| oedema | uncommon | | | | |
| malaise | | | | | frequency not known |
| Investigations | | | | | |
| hyperkalaemia | common | | uncommon† | common‡ | |
| increased alanine aminotransferase (ALT) § | rare | | | | |
| increase in blood urea, serum creatinine, and serum potassium | | | common | | |
| hyponatraemia | | | | | frequency not known |
| hypoglycaemia | | | | common | |

*Including swelling of the larynx, glottis, face, lips, pharynx, and/or tongue (causing airway obstruction); in some of these patients angioedema had been reported in the past in connection with the administration of other medicines, including ACE inhibitors
**Including Henoch-Schönlein purpura
‖ Especially in patients with intravascular depletion, e.g. patients with severe heart failure or under treatment with high dose diuretics
†Common in patients who received 150 mg losartan instead of 50 mg
‡reported in type 2 diabetic patients with nephropathy
§Usually resolved upon discontinuation

## OVERDOSAGE

With regard to over dosage in humans, limited data is available. Hypotension and tachycardia would be the most likely manifestation of over dosage; bradycardia could occur from parasympathetic (vagal) stimulation. Supportive treatment should be instituted if symptomatic hypotension occurs.

Through hemodialysis neither losartan nor the active metabolite can be removed.

Measures are depending on the time of medicinal product intake and kind and severity of symptoms. Stabilization of the cardiovascular system should be given priority. After oral intake, the administration of a sufficient dose of activated charcoal is indicated. Afterwards, close monitoring of the vital parameters should be performed. Vital parameters should be corrected if necessary.

STORAGE
Store below 30°C, protected from moisture

PACKING
Blister strips of 10 x 10’s and 3 x 10’s

Date of revision: July 2021

*Product Registration Holder/ Manufactured by:*
RANBAXY (MALAYSIA) SDN.BHD.
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